Orbit Commons — Visual Product and Delivery Plan

Status: Fictional concept for review. No product implementation is implied.
Audience: Product, design, engineering, operations, and pilot workspace managers.
Purpose: Help people find, reserve, and use shared rooms, desks, and equipment.

All policies below are proposed assumptions. All names, dates, metrics, scores, and chart data are synthetic. Charts illustrate potential reporting views; they do not report measured product results. The roadmap uses an arbitrary start date.

1. Product frame

Decision	Proposed default	Review condition
Workspace	One organization owns its resources and memberships.	Confirm with pilot managers.
Roles	Members book; managers approve; owners manage access.	Review the role matrix before development.
Reservation	A booking covers one exclusive resource and a half-open time interval.	Group bookings are a later candidate.
Availability	Pending, confirmed, and checked-in bookings block overlapping intervals.	Enforce this rule in the database.
Approval	Selected equipment requires a manager decision within 30 minutes.	Pending requests reserve the interval until a decision or expiry.
Confirmation	A booking succeeds only after the write transaction commits.	A conflict preserves entered details and shows alternatives.
Retry	A submission carries a key scoped to workspace and member.	Repeated submissions return the original result.
Time	Store instants in UTC; show the workspace time zone.	The sample workspace displays Europe/London.
Attendance	Check-in is allowed only during a confirmed booking.	An unvisited booking becomes a no-show at its end.

Decision	Proposed default	Review condition
Delivery	Notifications use a transactional outbox and retries.	Delivery failure does not revoke a booking.
Scope	Search, booking, approval, check-in, cancellation, and reports.	Payments and recurring reservations are later candidates.

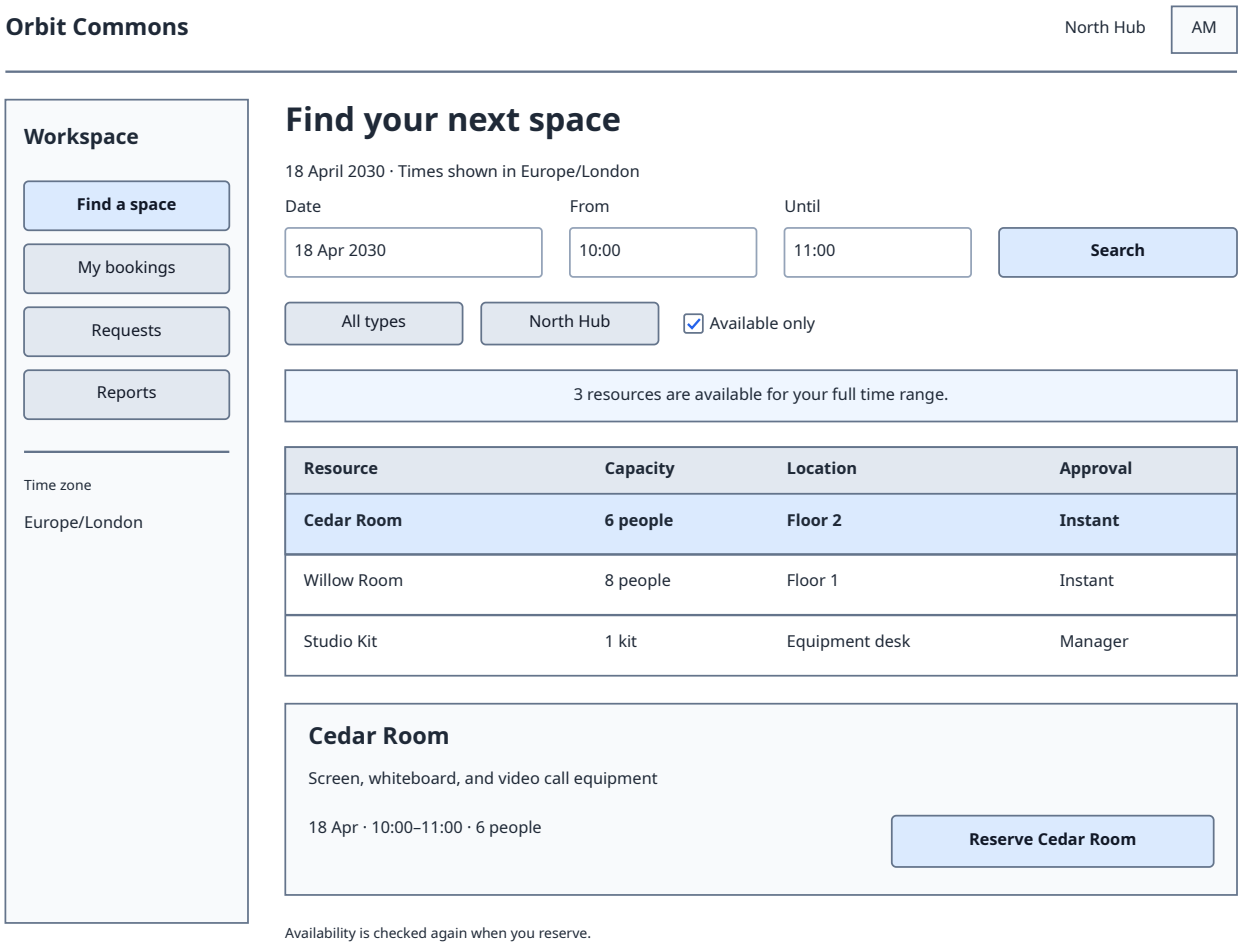
Role	Find availability	Own bookings	Approve requests	Manage resources	Manage access
Member	Yes	Create, check in, cancel before start	No	No	No
Manager	Yes	Yes	Yes	Yes	No
Owner	Yes	Yes	Yes	Yes	Yes

2. Screen prototypes

Each block shows one screen or one state. Controls describe intended behavior; these static prototypes are not an interactive application. Screen formats vary to demonstrate three supported UI renderers.

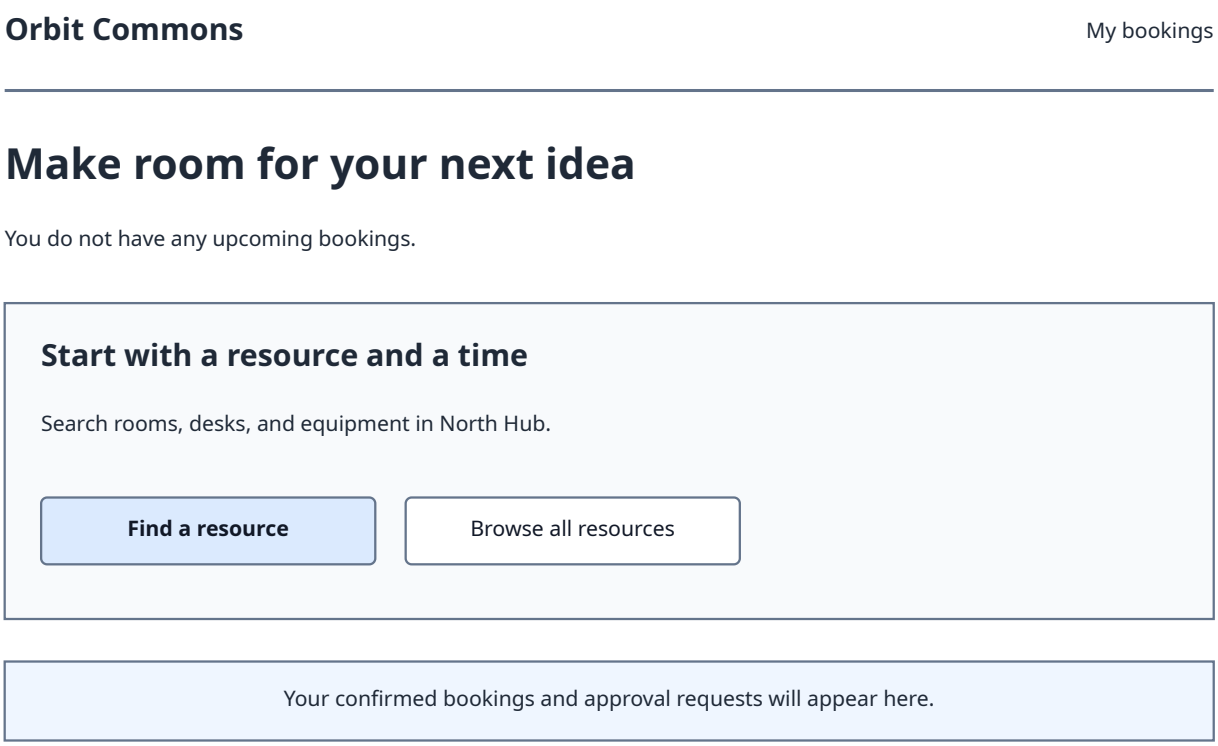
U01 — Find an available resource

Member desktop search. The selected resource has a clear reservation action and an explicit time zone.



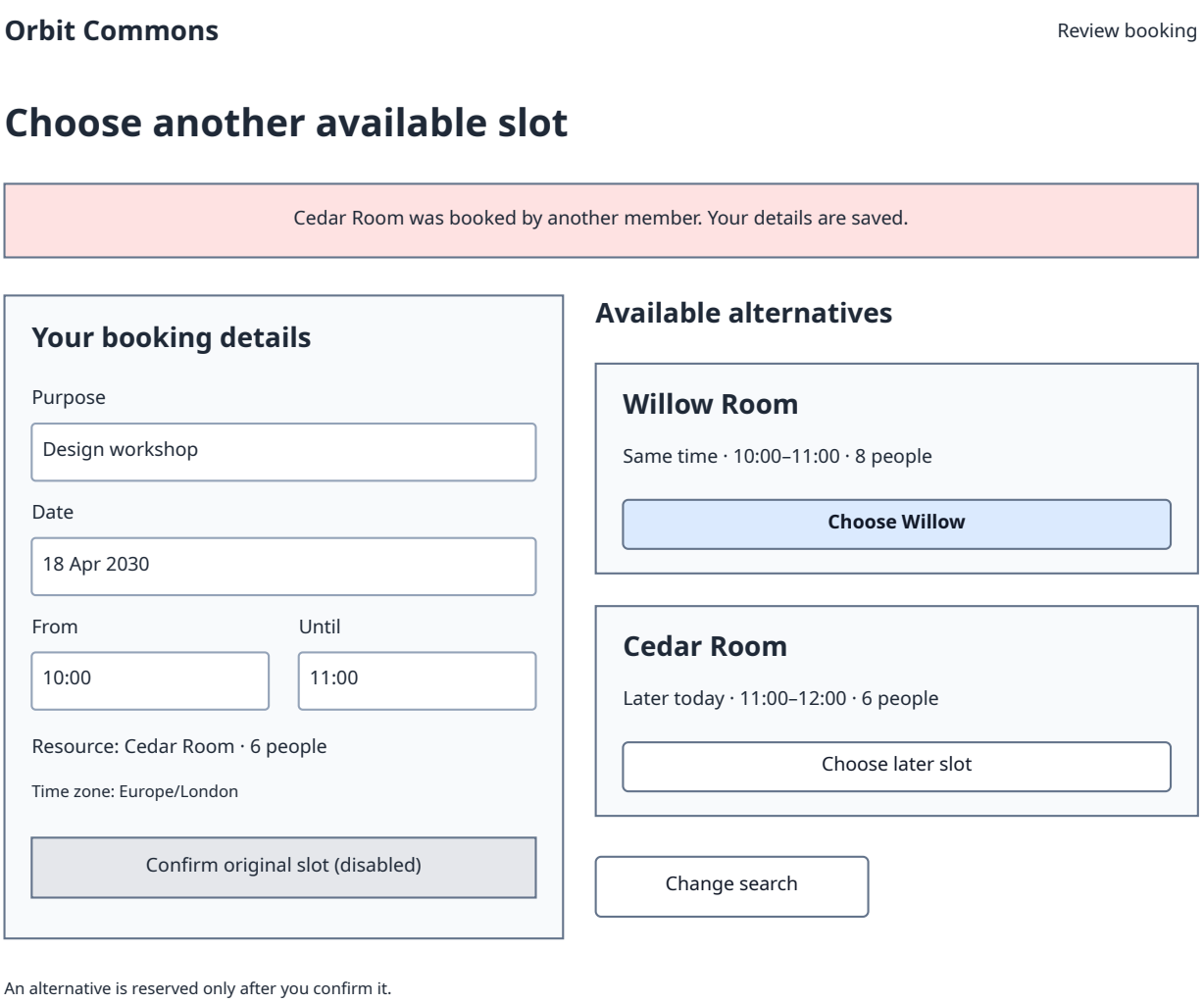
U02 — First booking: empty state

A first-time member receives a useful next step, a secondary browse action, and an explanation of what will appear here.



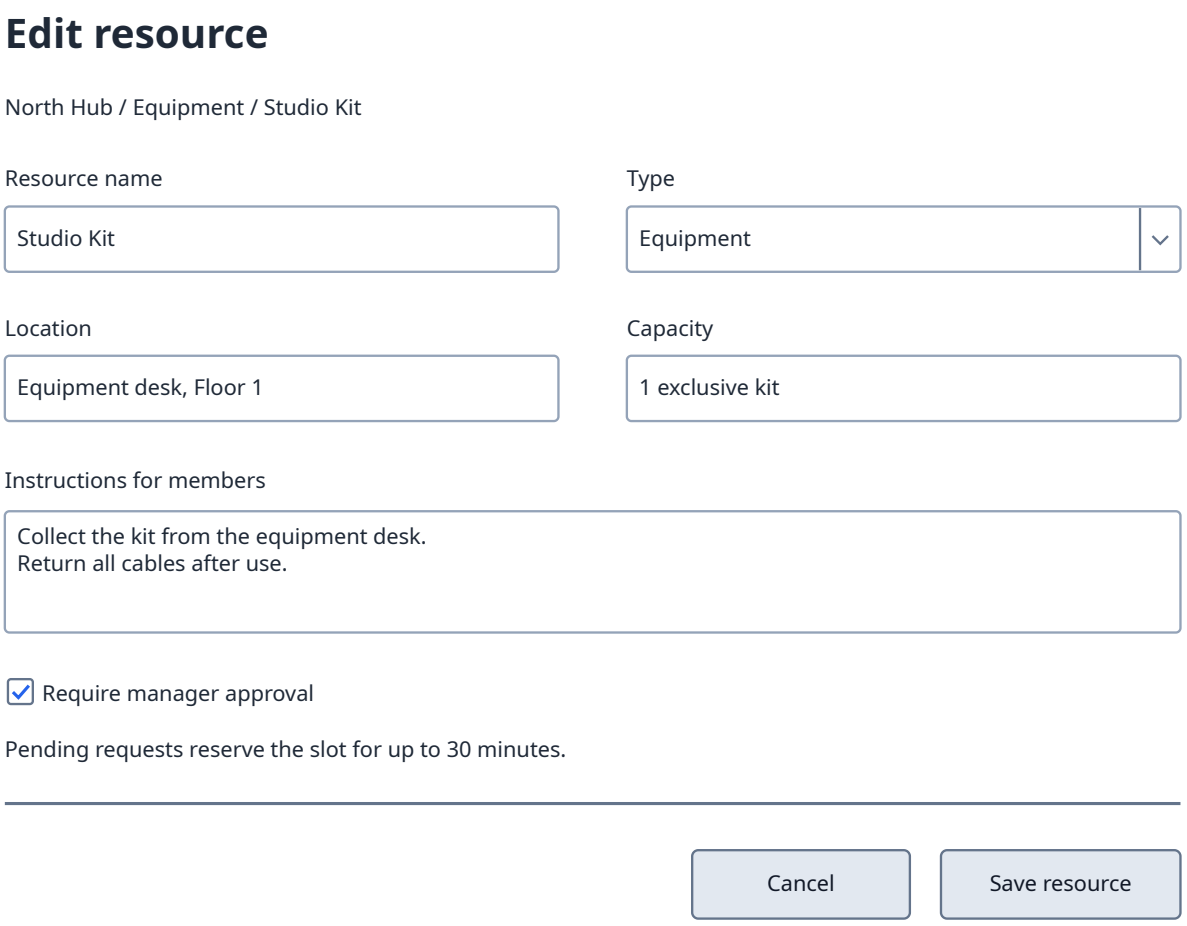
U03 — Booking conflict: recover without retyping

The conflict state preserves input, disables the stale action, and offers two concrete recovery paths.



U04 — Resource editor

Manager desktop editor. Approval is a visible resource policy; the form exposes the reservation effect of pending requests.



U05 — Manager approval review

A manager can inspect the purpose, see the expiry time, and submit an explicit approval or rejection.

Review equipment request

Pending approval · Expires at 09:35 Europe/London

Alex Morgan · Studio Kit

18 Apr 2030 · 10:00–11:00 · North Hub

Purpose: Record the team product demo.
The requested interval is reserved while this request is pending.

Decision note (required for rejection)

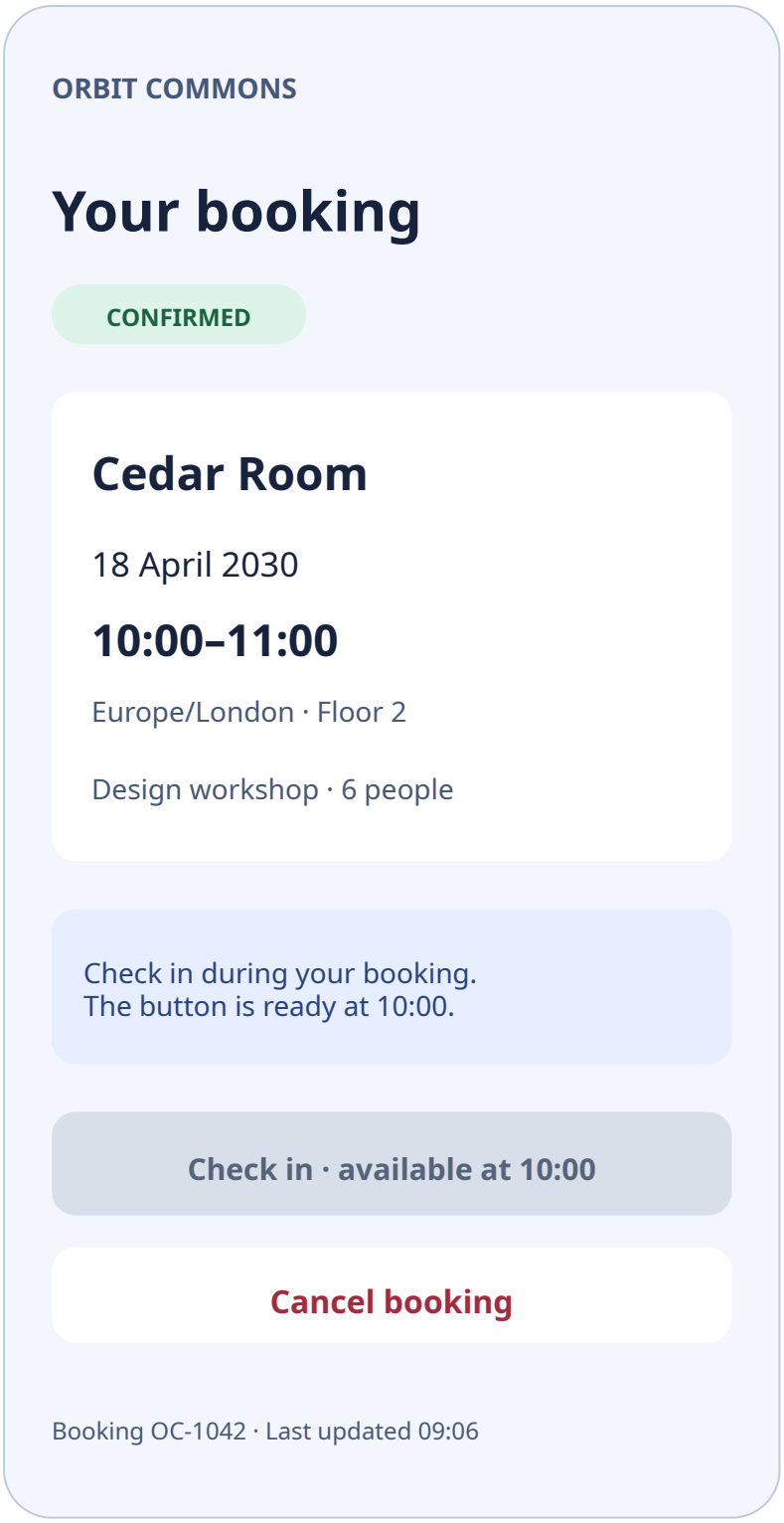
Equipment safety briefing completed.

Reject request | Approve booking

The decision is applied only if the request is still pending.

U06 — Mobile booking details before start

Mobile detail screen. Status, place, time, purpose, and cancellation are visible. Check-in is disabled before the booking starts.



U07 — Instant booking confirmation

Detailed confirmation state. A committed reservation is visibly confirmed; the next action opens its details.

ORBIT COMMONS / NORTH HUB

✓

Your room is reserved

You can find this booking in My bookings.

Cedar Room

18 April 2030 · 10:00–11:00 · Europe/London

Floor 2 · 6 people · Video call equipment

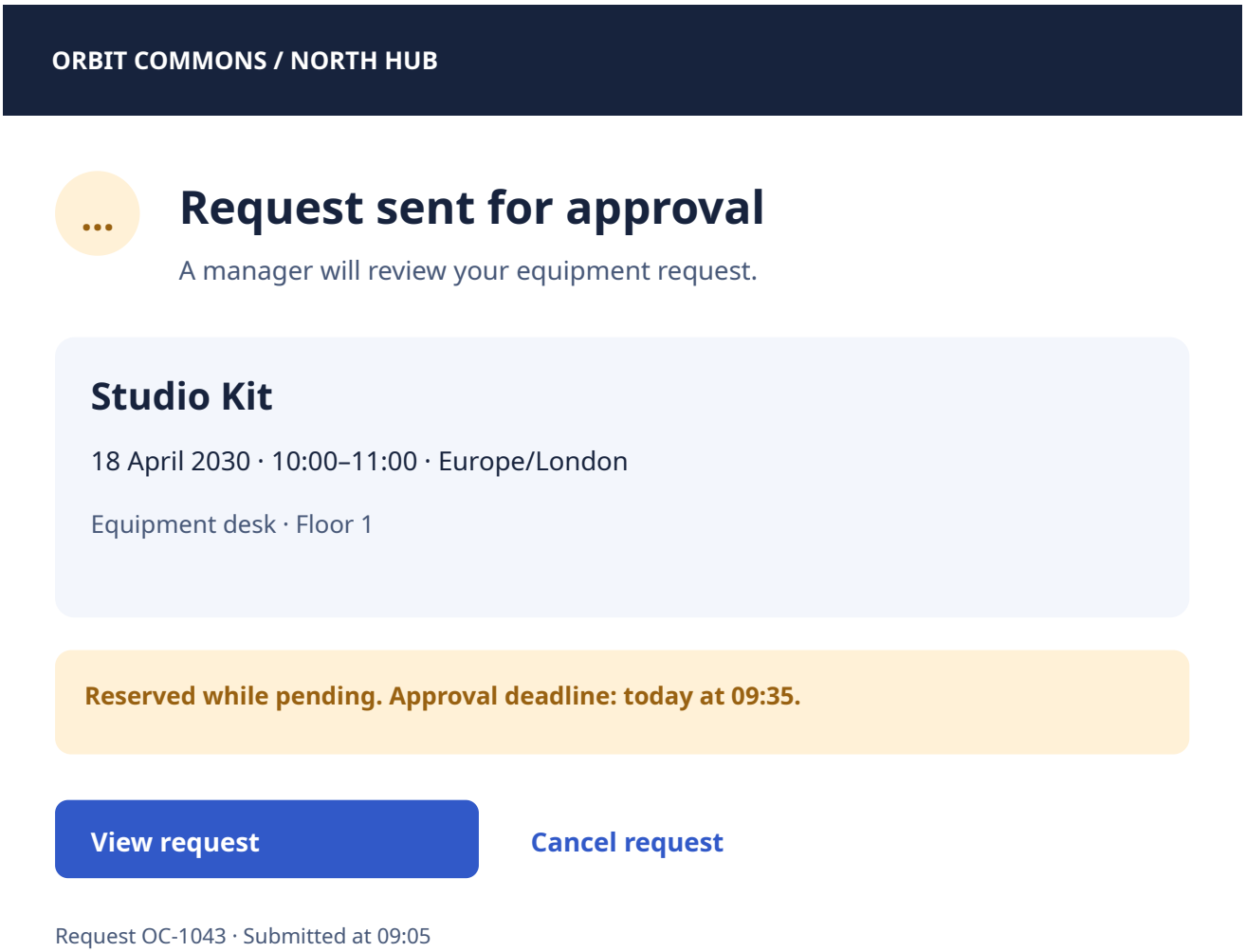
Confirmed. Check in when your booking starts at 10:00.

View booking **Find another resource**

Booking OC-1042 · Confirmation sent to your inbox

U08 — Approval request submitted

Equipment uses a different result state: pending approval, a clear deadline, and a cancel action. Submission is not presented as final approval.

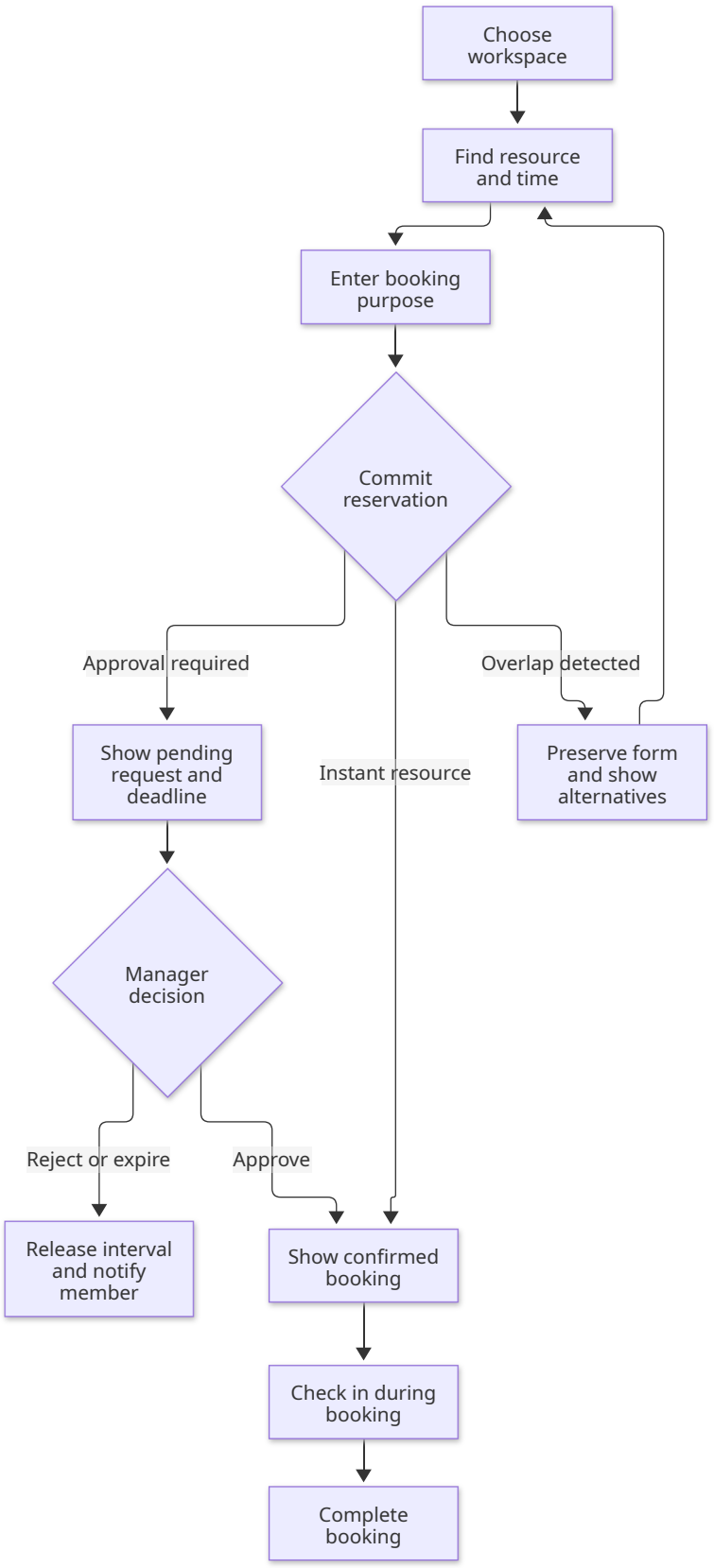


3. Product flows and system design

The following diagrams propose a small service architecture. A workspace identifier accompanies every request and event. The database is the authority for overlapping reservations.

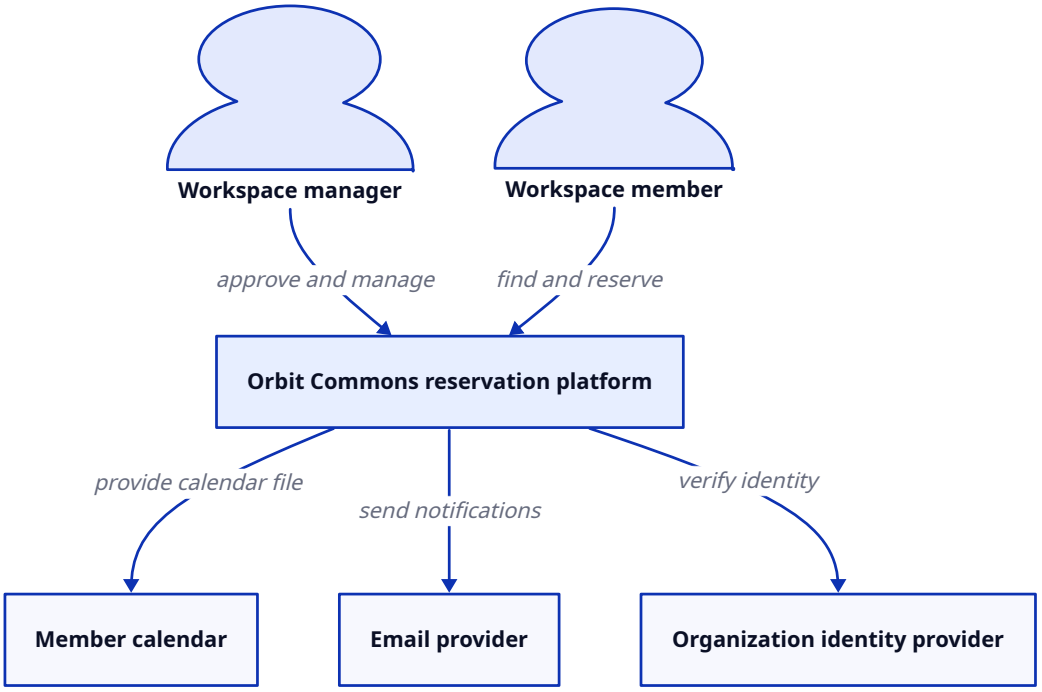
D01 — Member booking journey

The main task includes successful booking, concurrent conflicts, and manager approval.



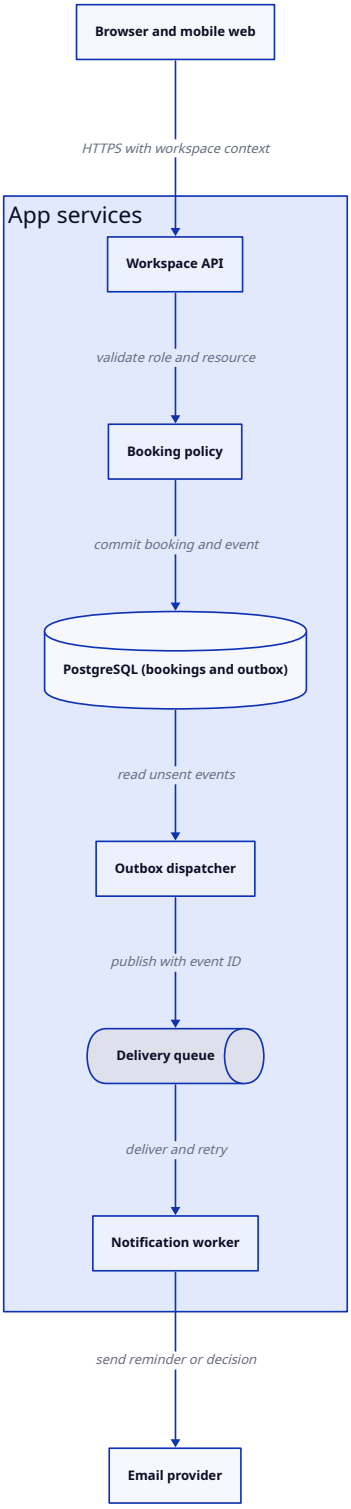
D02 — System context

People and external boundaries. Calendar support starts with a downloadable file; provider synchronization is a later option.



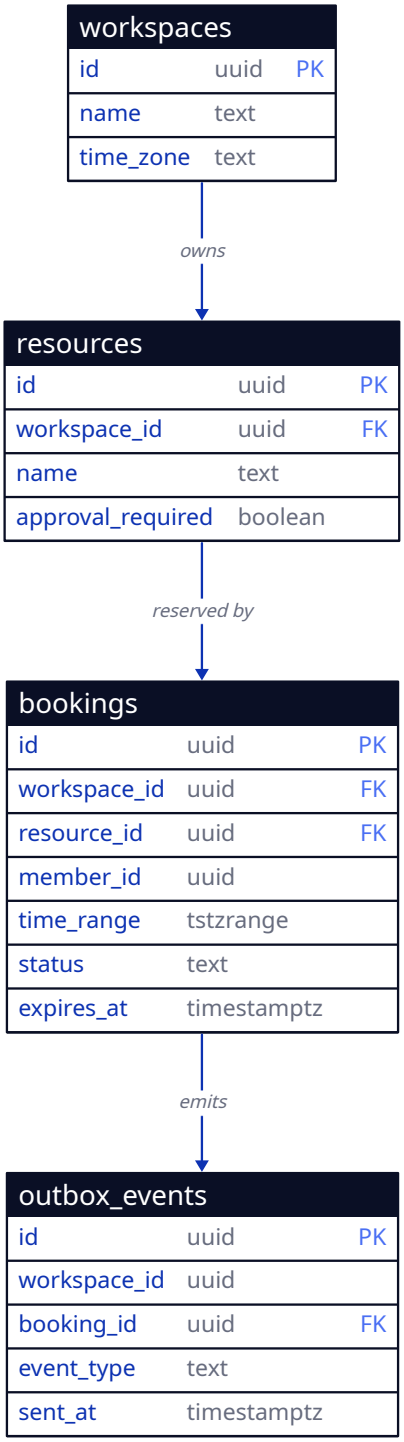
D03 — Runtime containers and write boundary

Booking and outbox records share a transaction. A queue failure is retried without creating another reservation.



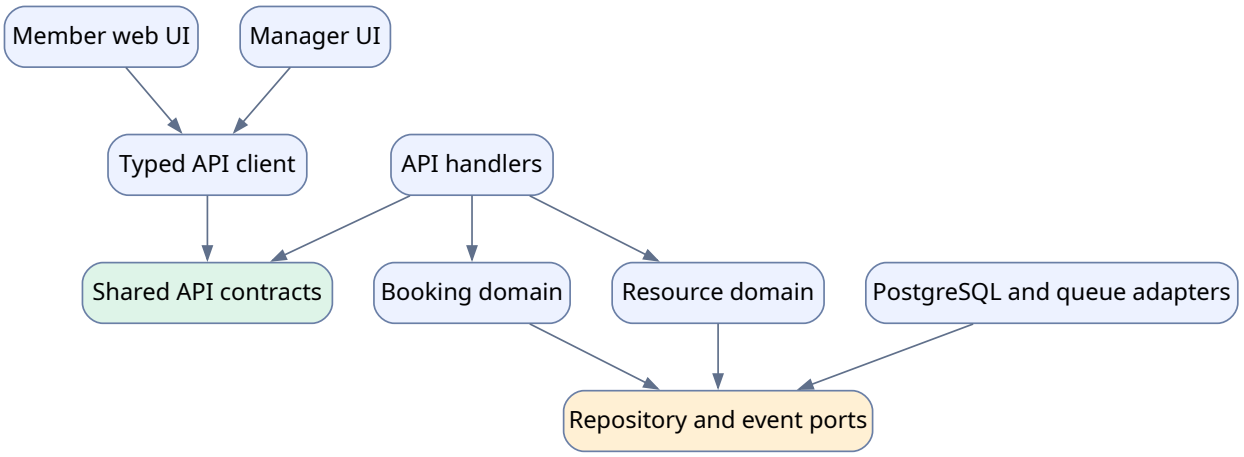
D04 — Reservation storage model

Tables show the reservation write path. Add a database exclusion constraint on workspace, resource, and overlapping time ranges for blocking statuses.



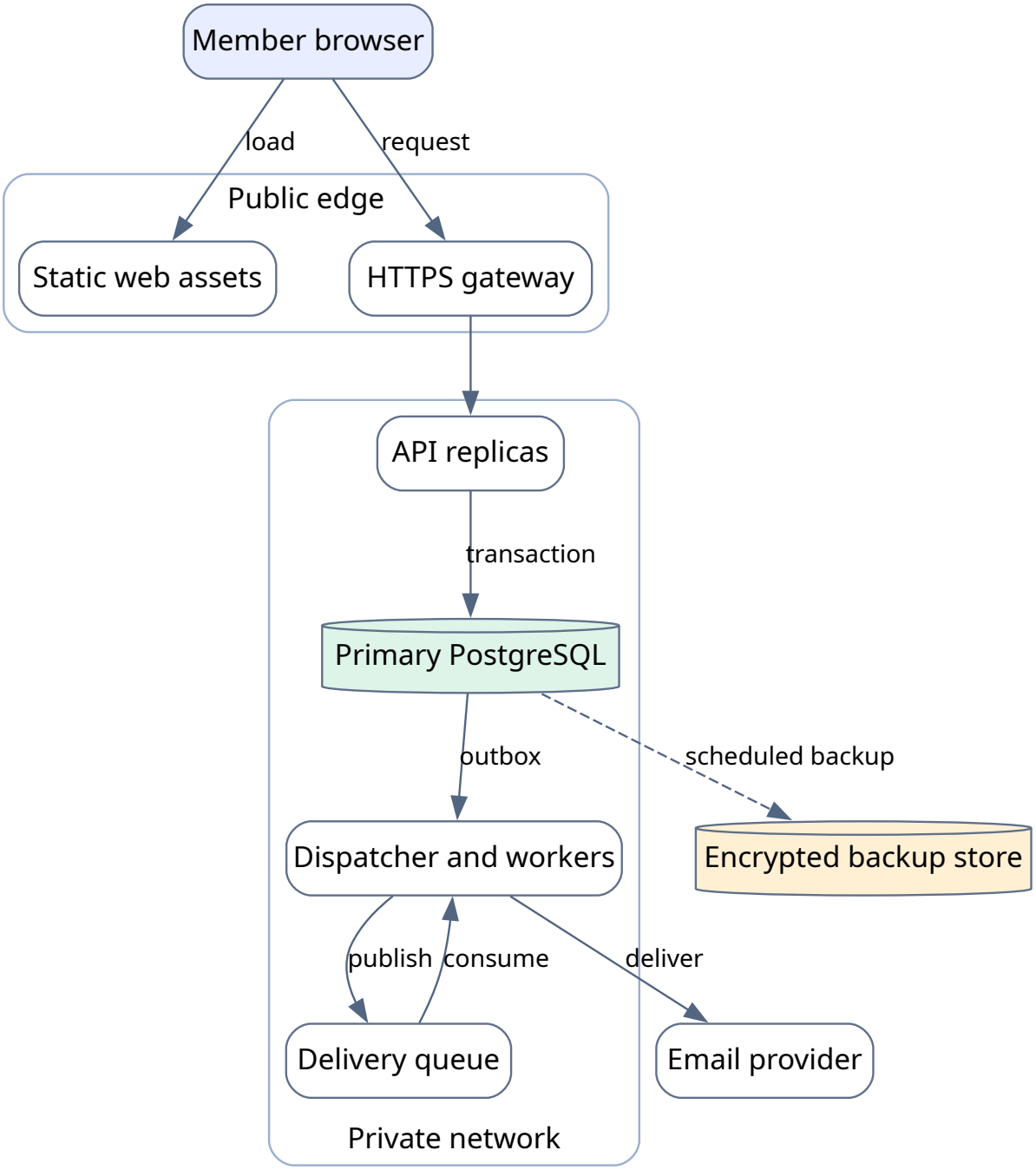
D05 — Module dependencies

Arrows mean “depends on.” Domain modules use ports; storage and queue adapters implement those ports.



D06 — Deployment zones

A proposed single-region topology. Restore drills and worker queue recovery are release criteria; multi-region operation is outside the first pilot.



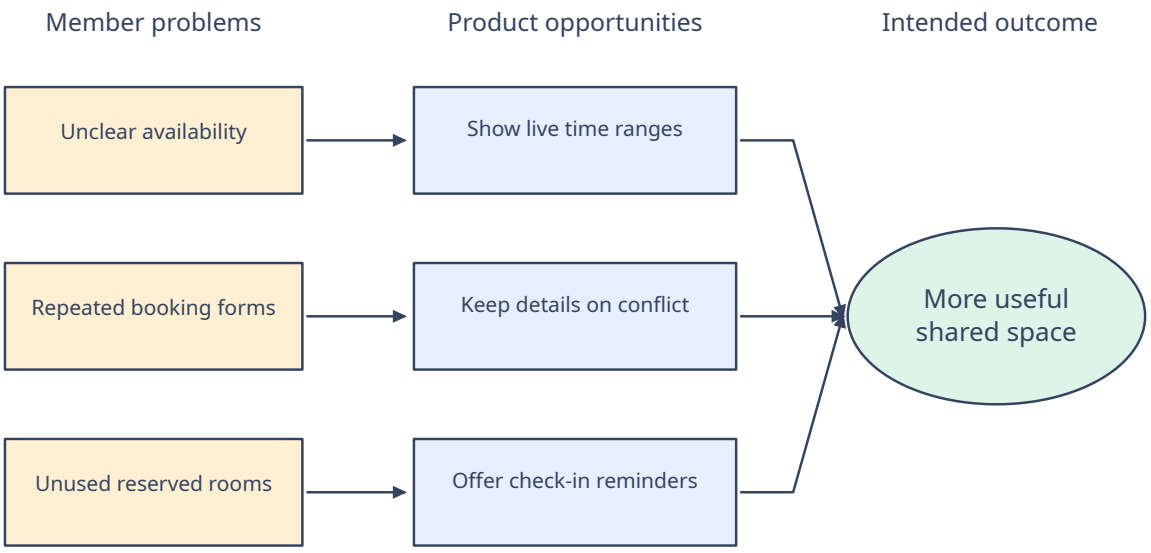
D07 — Reliable booking event delivery

Structural draw.io JSON groups the transaction and delivery boundaries. The renderer chooses node positions automatically.



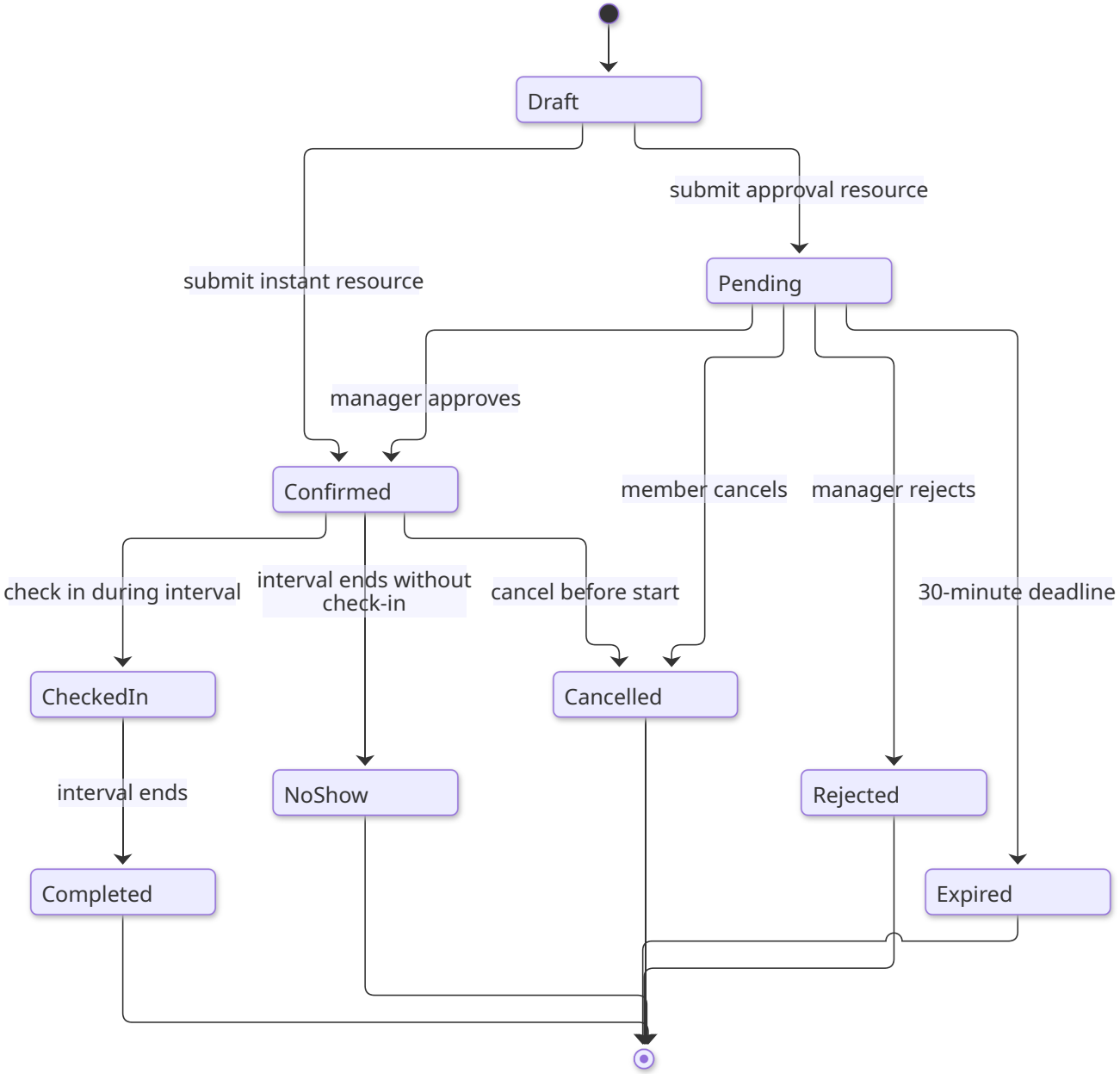
D08 — Problem and opportunity map

A positioned discovery sketch connects three member problems to proposed product responses. Arrows express a hypothesis, not measured causation.



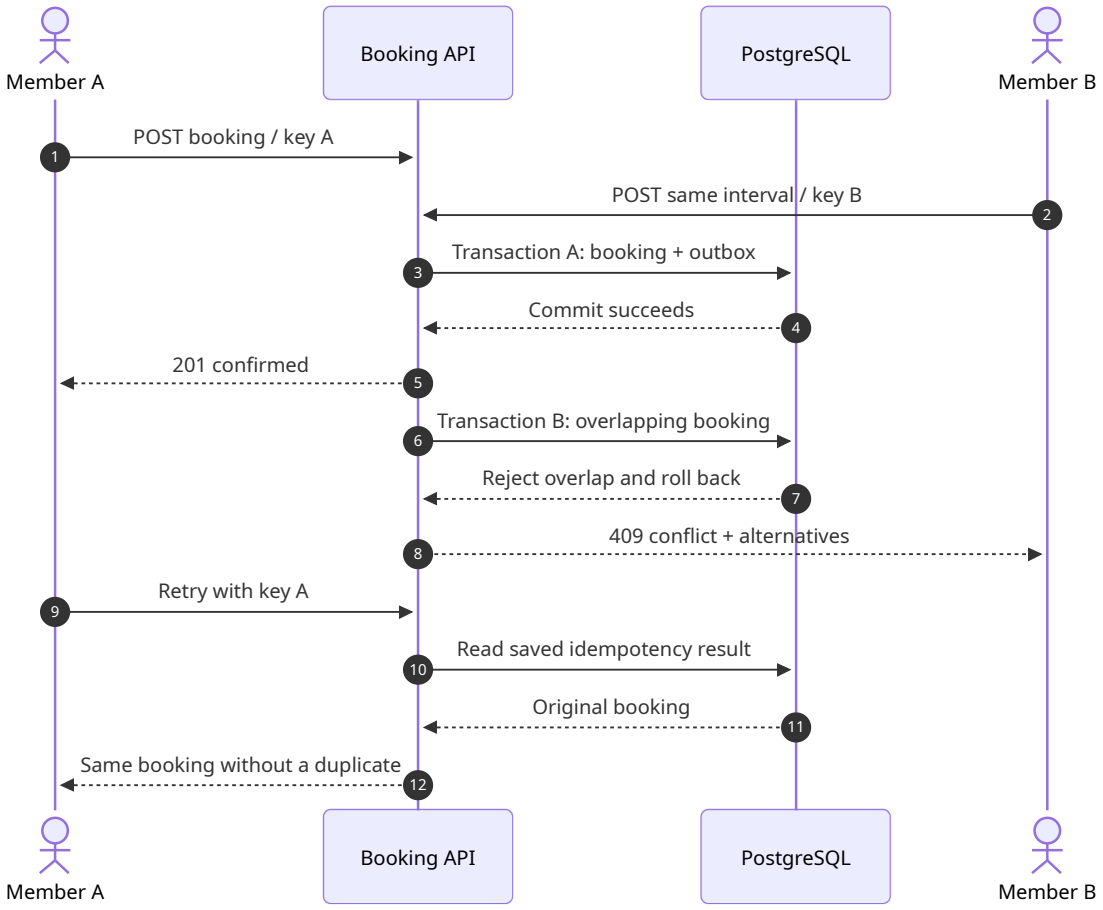
D09 — Booking lifecycle

Draft is local form state. Pending, confirmed, and checked-in states block overlaps. Terminal states release the reservation; time ranges are still enforced for active states.



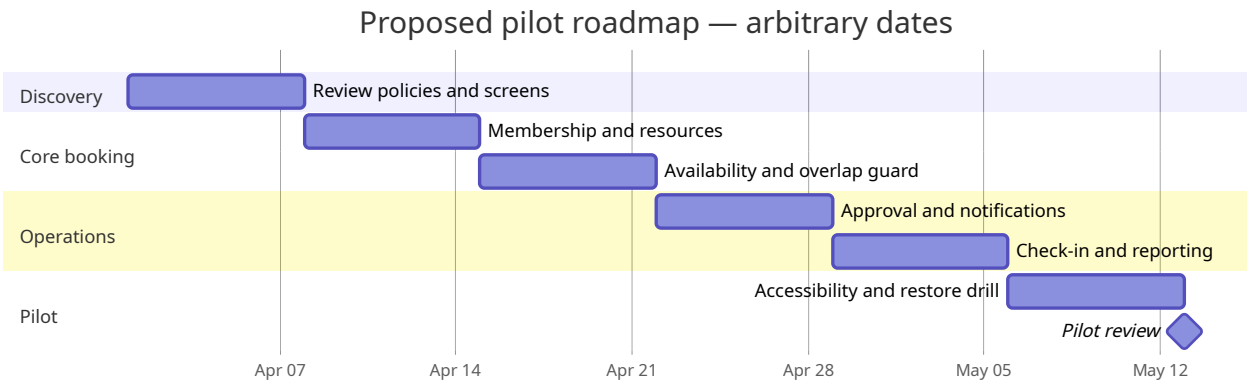
D10 — Concurrent submissions for one resource

The database selects the winner. A retry returns the stored result. A losing request produces neither a booking nor an outbox event.



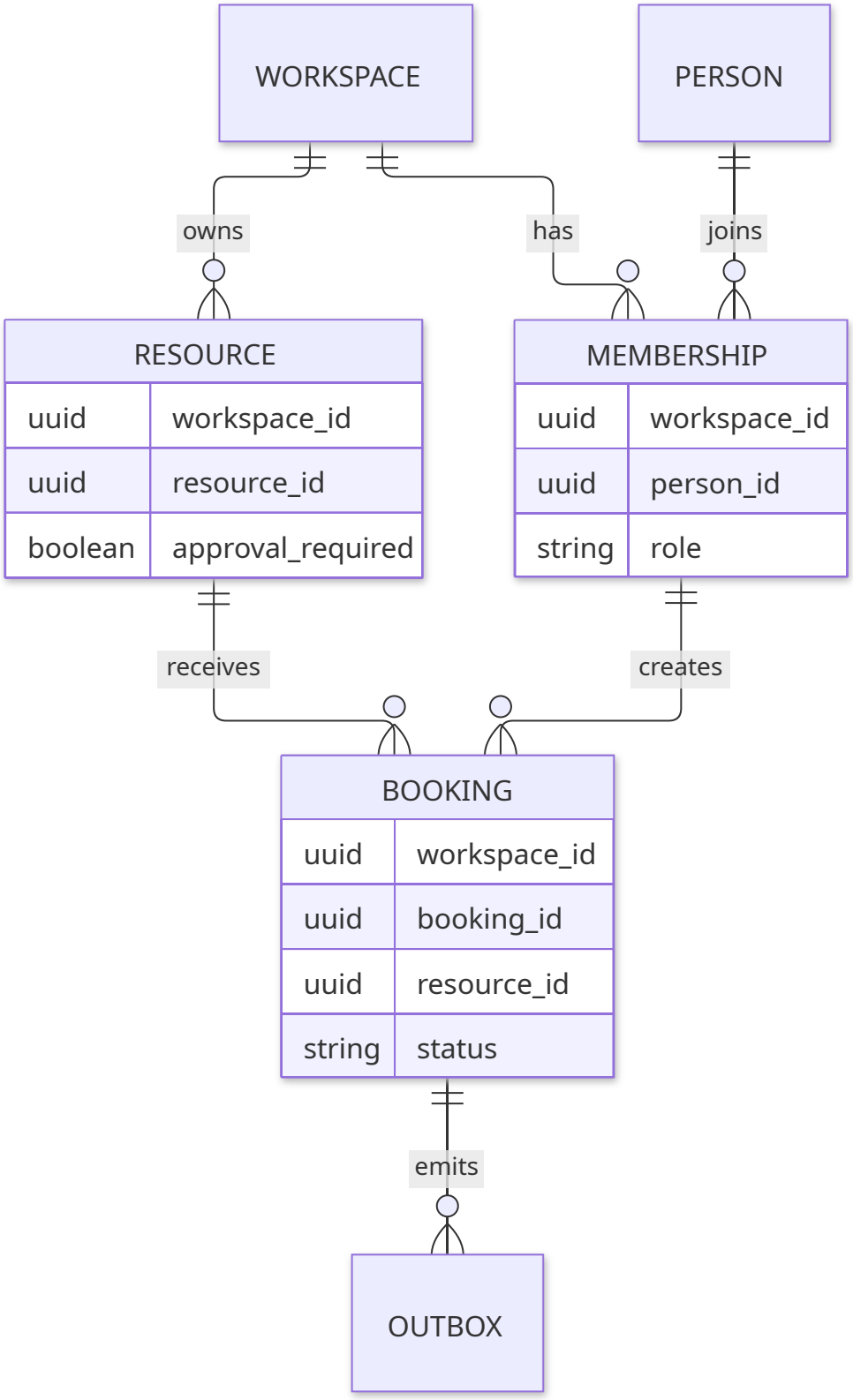
D11 — Illustrative six-week delivery roadmap

Sequential work packages make dependencies visible. Dates are a demonstration schedule, not a delivery commitment.



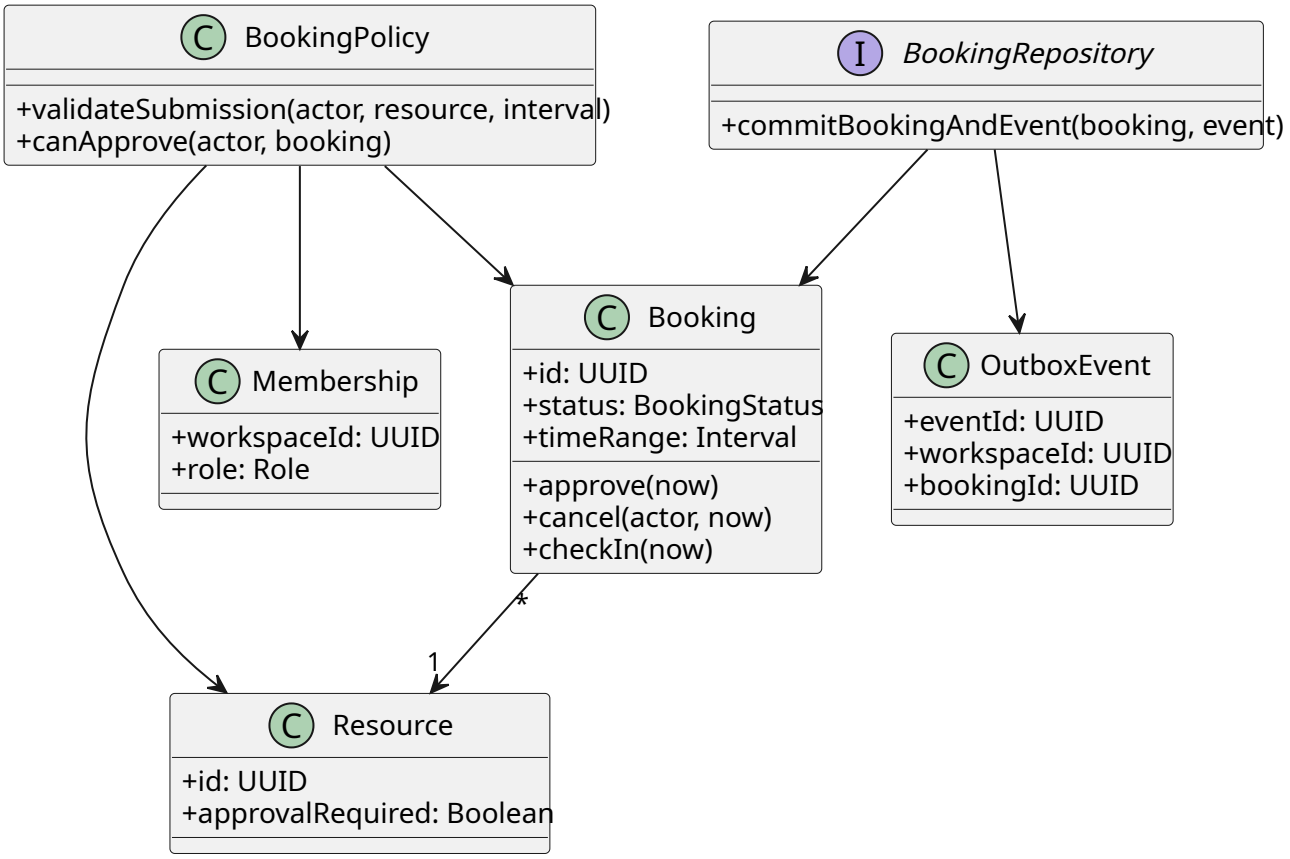
D12 — Membership and reservation cardinalities

A person can belong to several workspaces. Reservation relationships must preserve the same workspace across membership, resource, booking, and event.



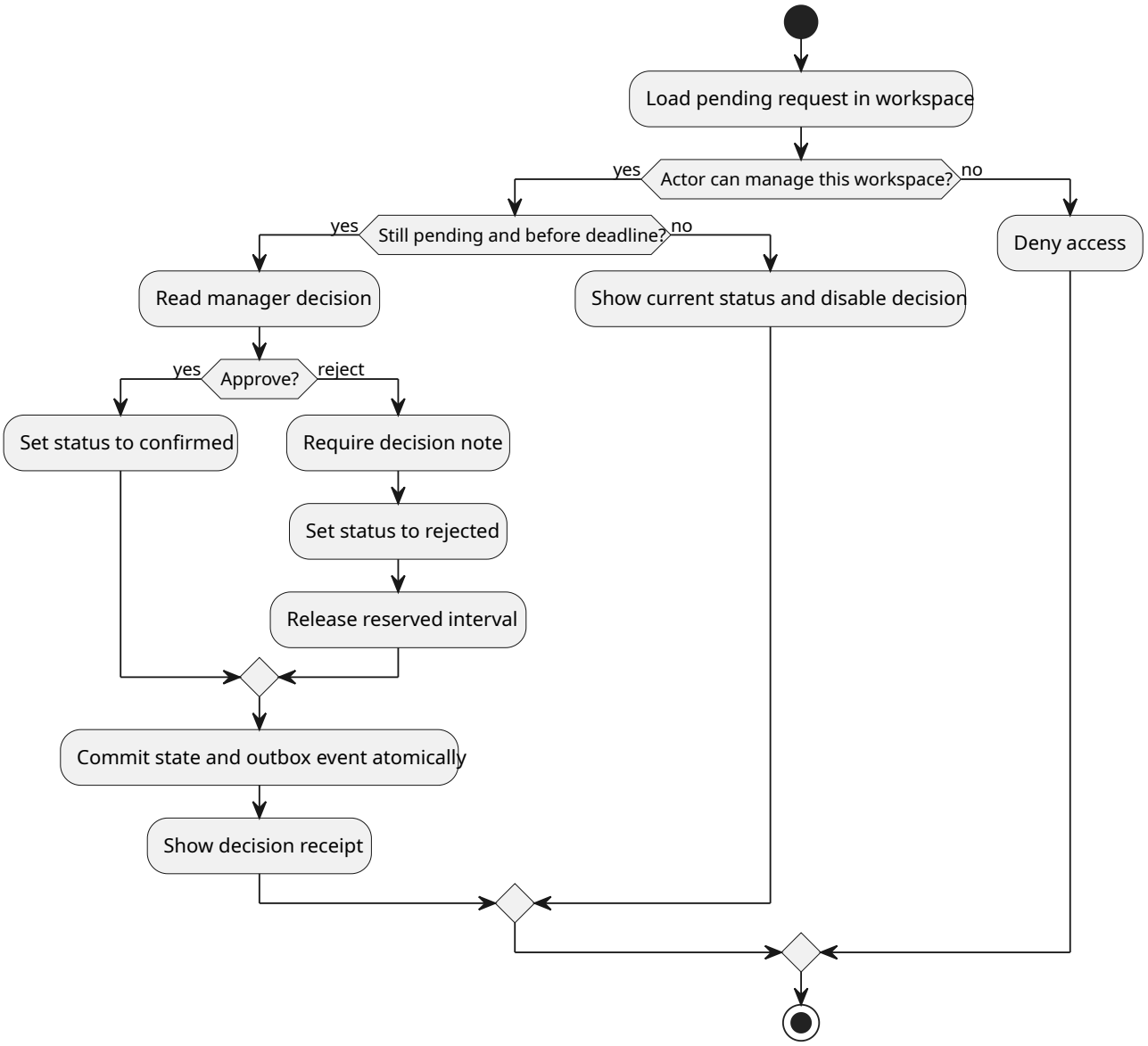
D13 — Domain responsibilities

The booking owns state transitions. A policy checks access and resource rules; one repository operation commits the booking and its event.



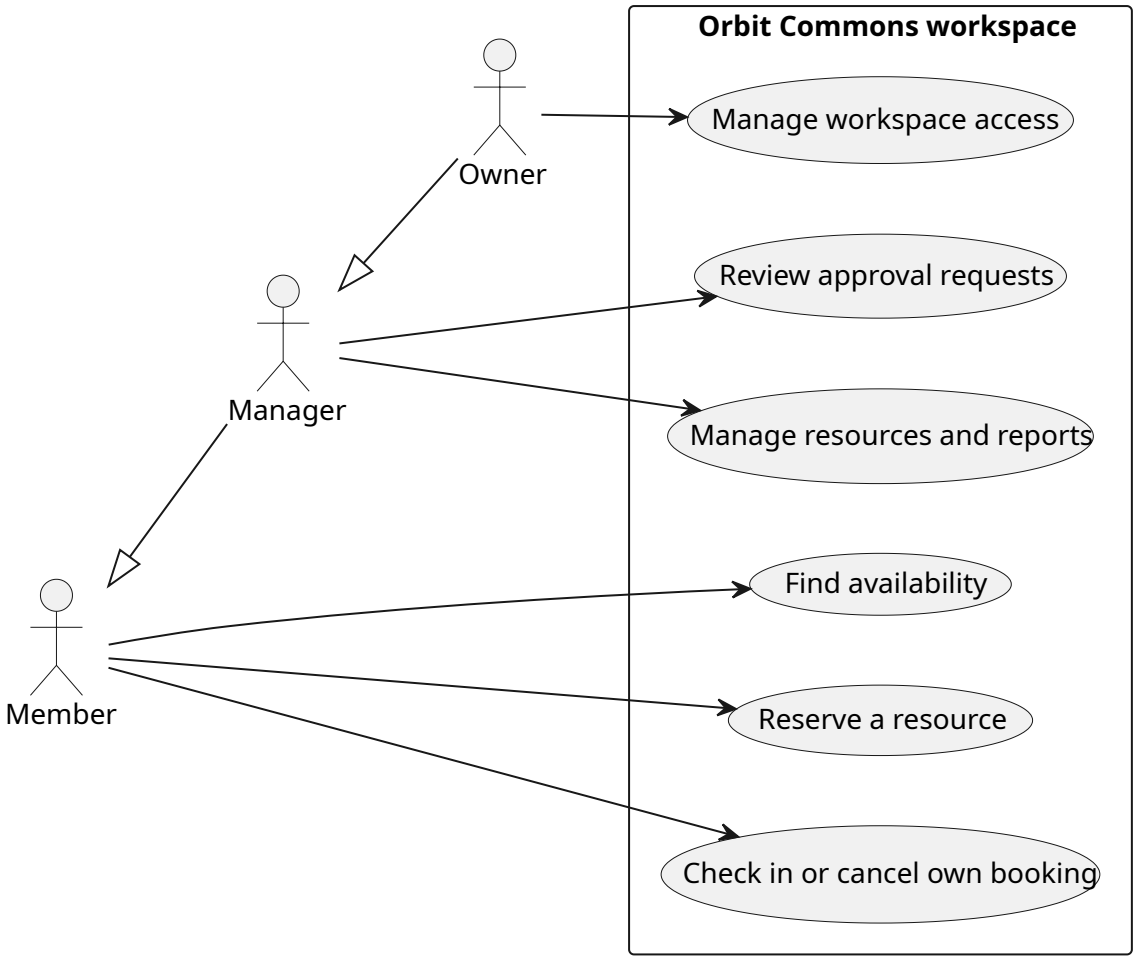
D14 — Approval decision activity

The write rechecks status and expiry inside the transaction. A concurrent expiry or earlier decision makes a stale approval action fail safely.



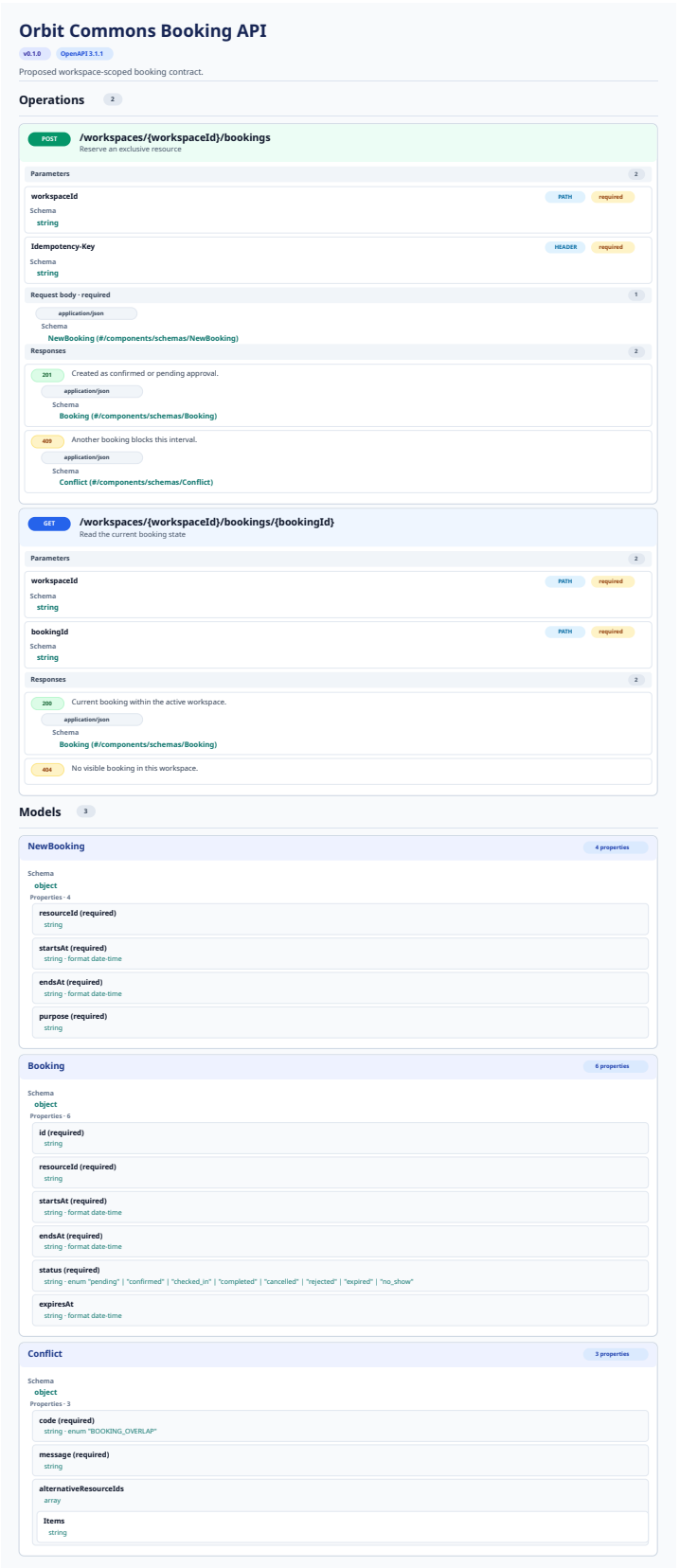
D15 — Role use cases

Manager and owner inherit member capabilities. All actions are scoped to the actor’s active workspace membership.



D16 — Booking API contract slice

A compact write/read slice. Every operation requires authenticated workspace membership. Approval, check-in, cancellation, expiry, and authentication endpoints need a complete contract before implementation.



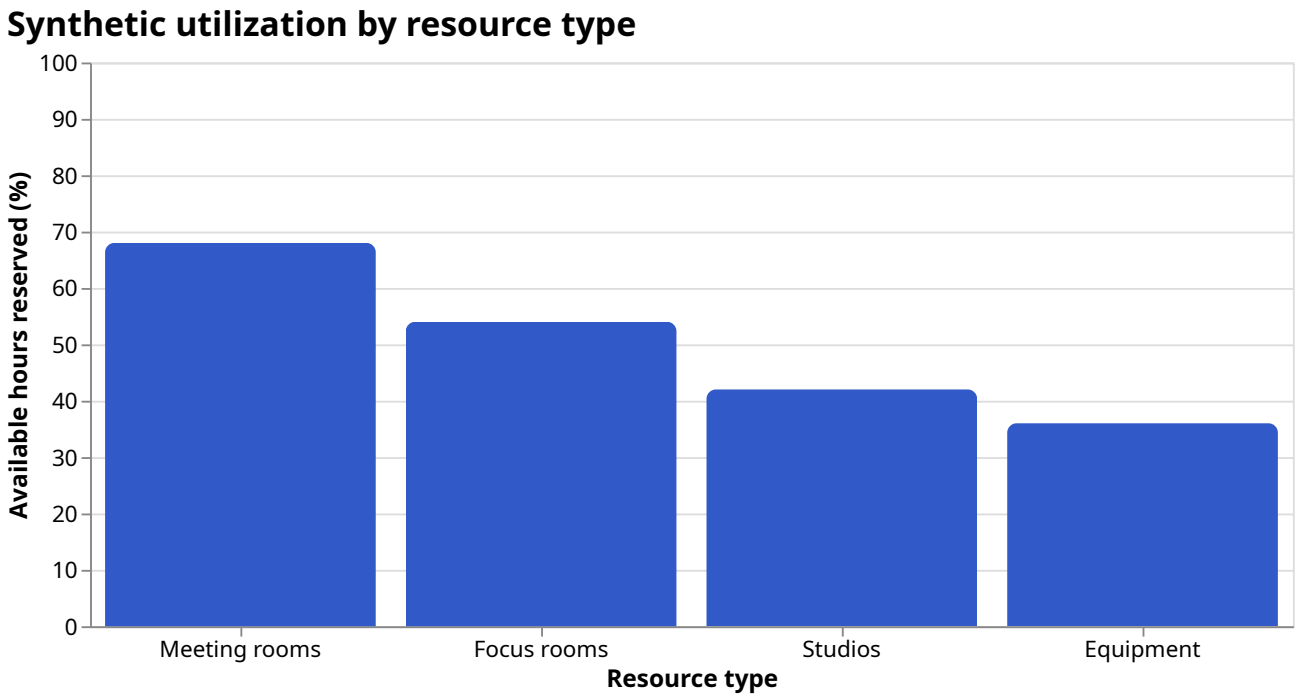
4. Analytics and operational chart gallery

Every chart uses embedded synthetic rows. Views are separate so each chart can be reviewed or exported on its own. Reporting must apply workspace access rules before aggregation.

Metric	Proposed definition	Interpretation limit
Resource utilization	Reserved hours / published available hours.	Exclude closed hours from the denominator.
Booking volume	New bookings created in a workspace and period.	A booking is counted once; delivery retries do not add volume.
Attendance rate	Checked-in eligible bookings / eligible bookings.	Define the observation window before using this metric.
Booking lead time	Booking start instant minus creation instant.	Sample distributions below are illustrative.
Request latency	End-to-end request duration in milliseconds.	Raw samples and p95 summaries use different charts.
Release coverage	Accepted criteria / agreed criteria.	Example scores do not indicate implementation progress.

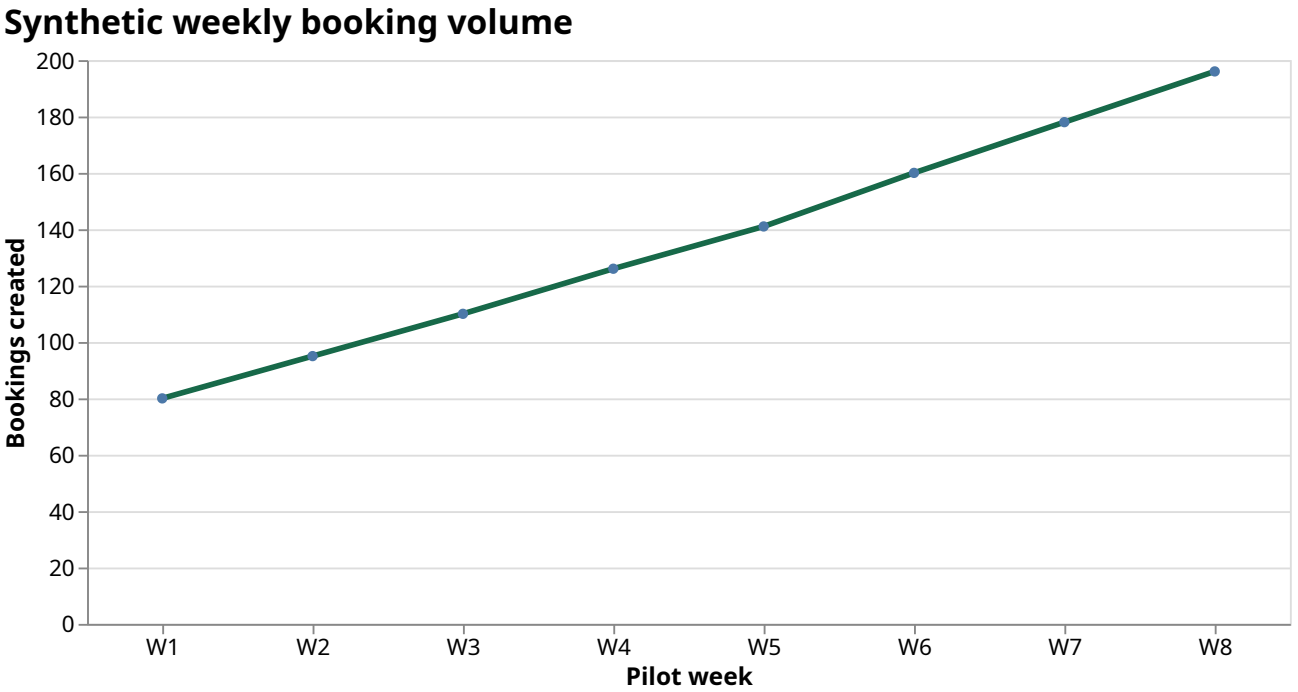
C01 — Bar chart: utilization by resource type

Compare utilization levels. Percentages use the same illustrative reporting period and denominator definition.



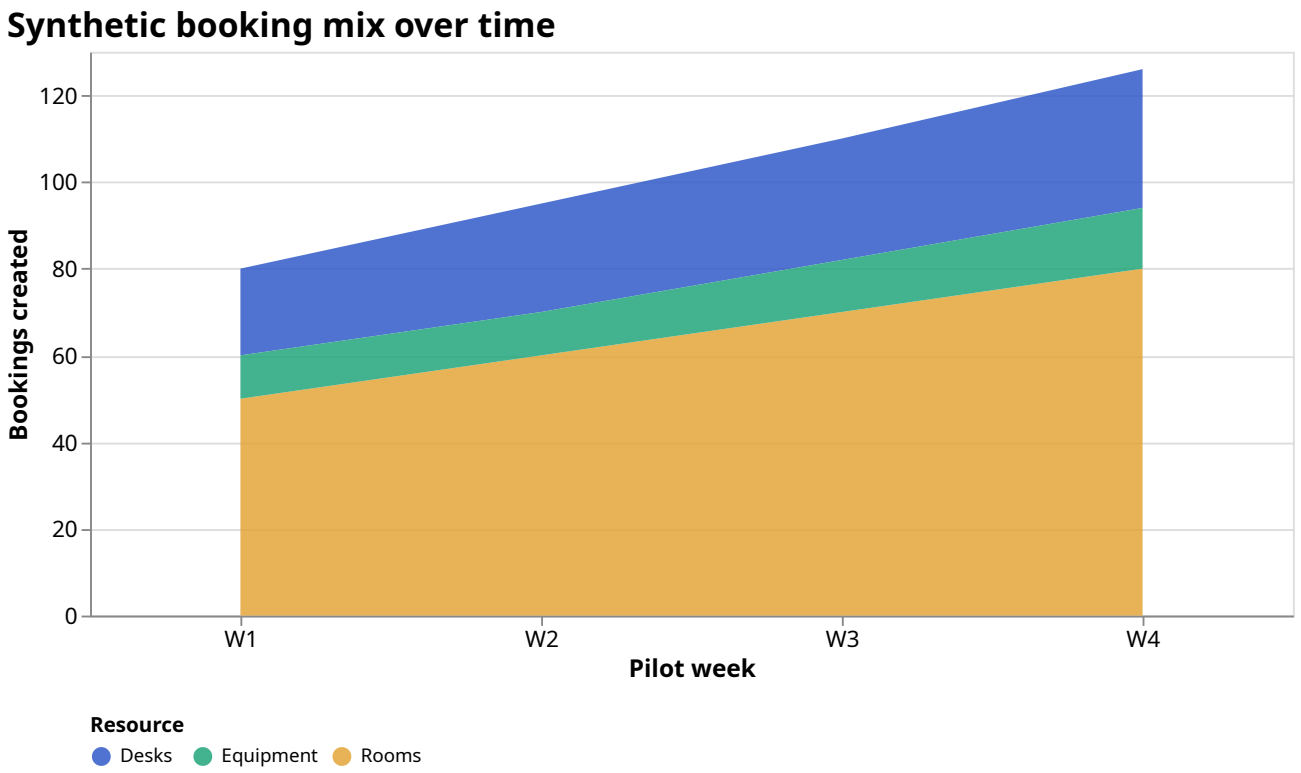
C02 — Line chart: weekly booking volume

Show a trend without implying a forecast. Values are scenario data, not observed adoption.



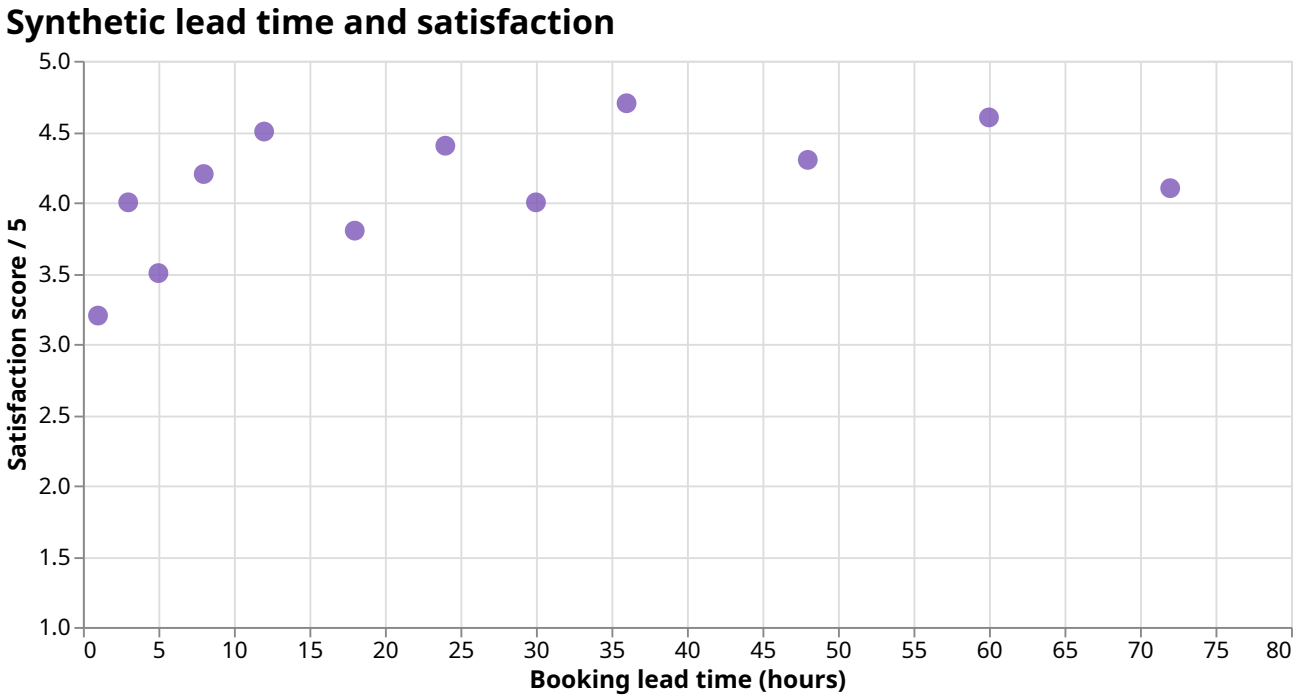
C03 — Stacked area chart: changing booking mix

A stacked area view shows composition and total volume. Weekly totals match W1–W4 in C02.



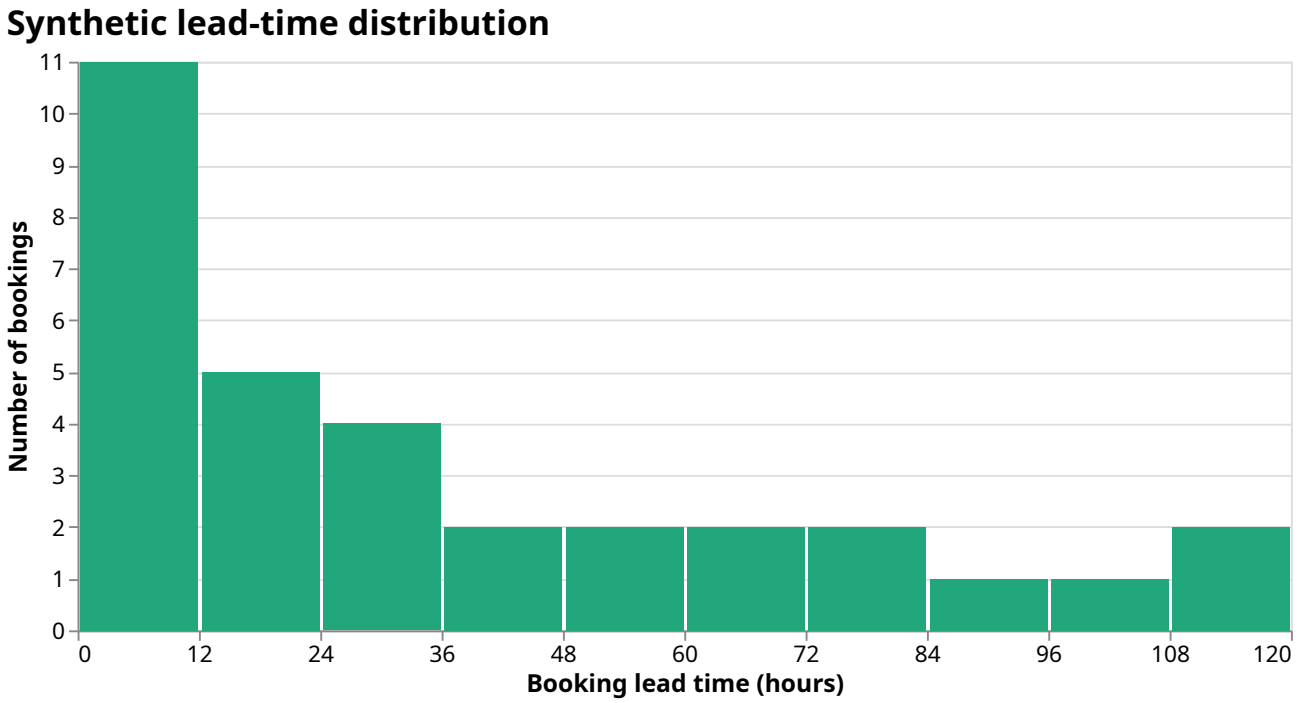
C04 — Scatter plot: lead time and satisfaction

Each point represents a fictional survey response. A visible association would not establish causation.



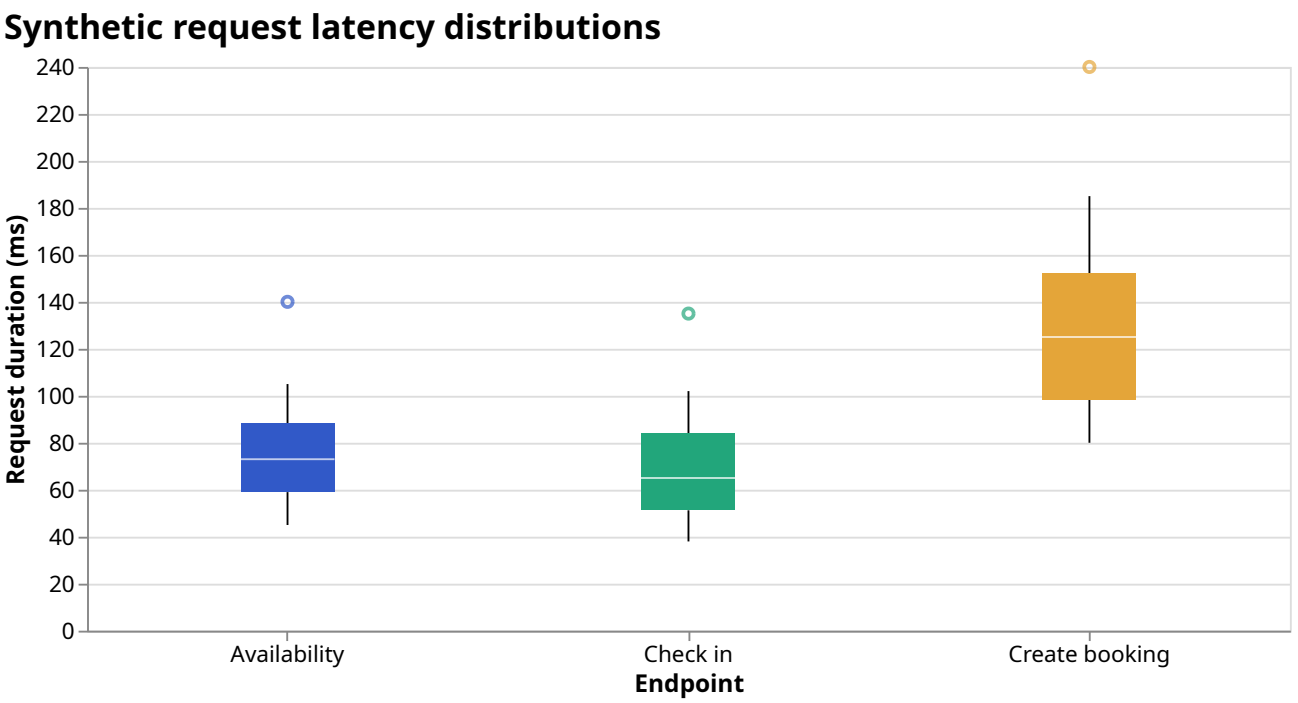
C05 — Histogram: distribution of booking lead time

A true histogram bins a continuous variable into 12-hour intervals. This differs from the categorical bar chart in C01.



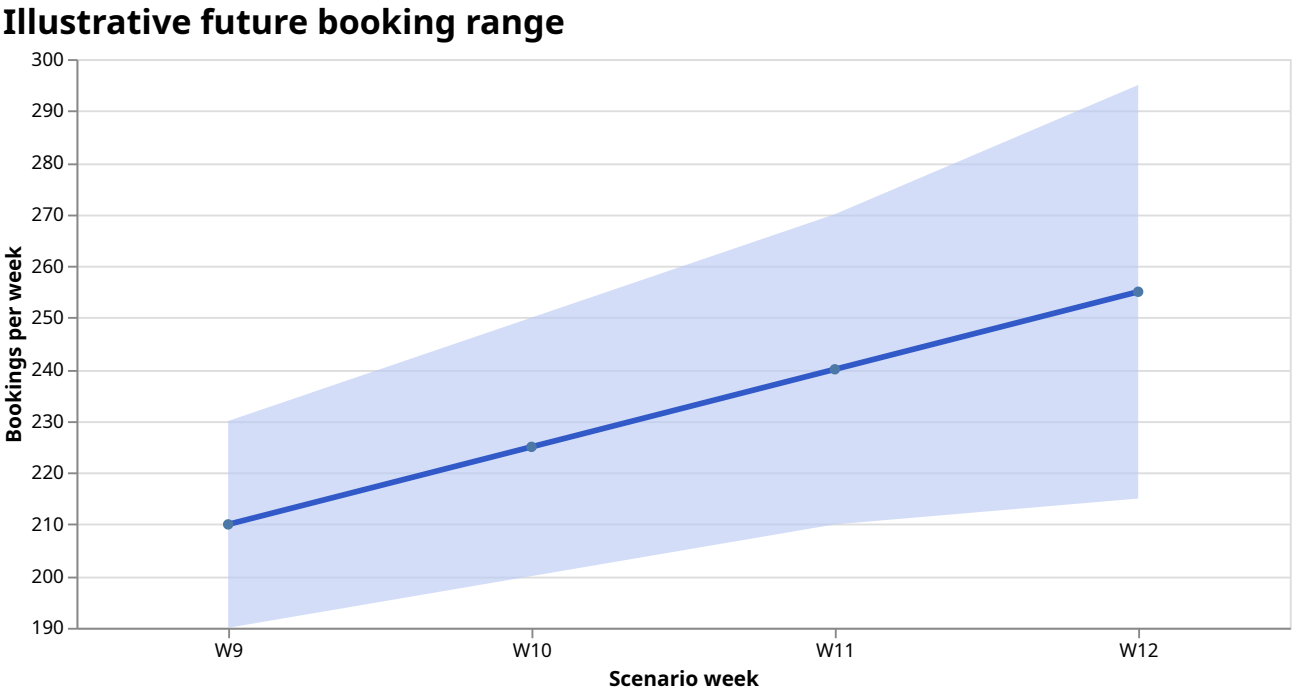
C06 — Box plot: request latency spread

Raw request samples show medians, interquartile ranges, whiskers, and outliers. These values are not p95 aggregates.



C07 — Layered range chart: a capacity scenario

The band is a manually chosen planning range around a central scenario. It is not a statistical confidence interval.



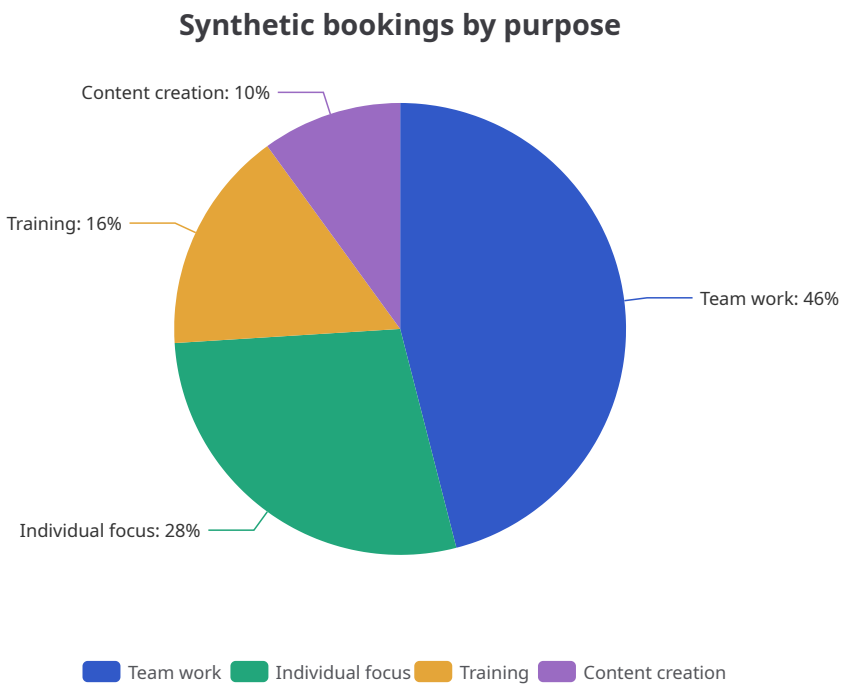
C08 — Mixed chart: bookings against a plan

Bars and a dashed line share one unit. Booking values match W5–W8 in C02; the target is a separate illustrative plan.



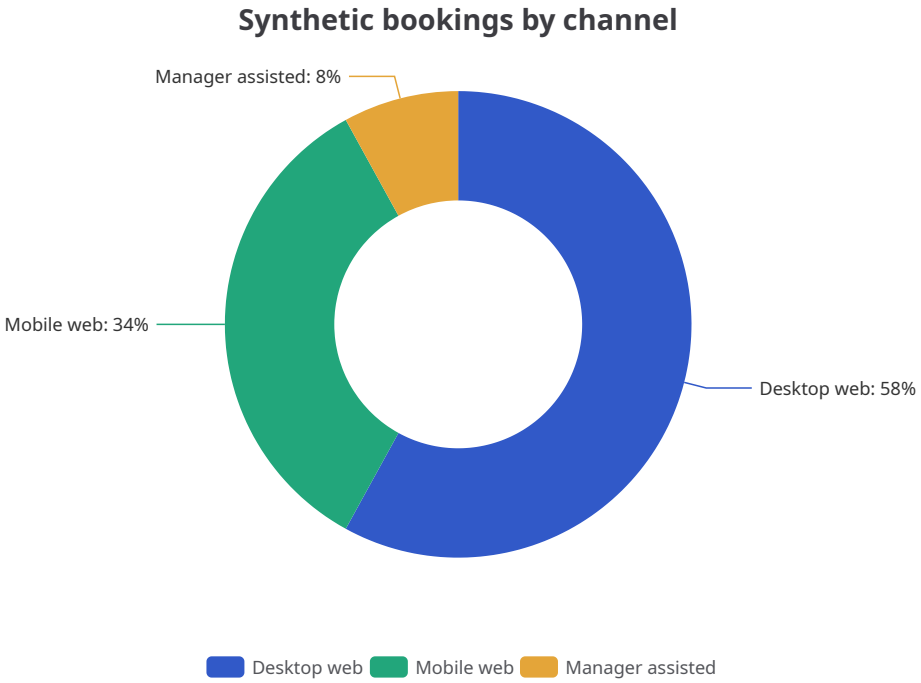
C09 — Pie chart: booking purpose

A conventional pie chart shows a small set of categories that total 100%. It is separate from the ring chart below.



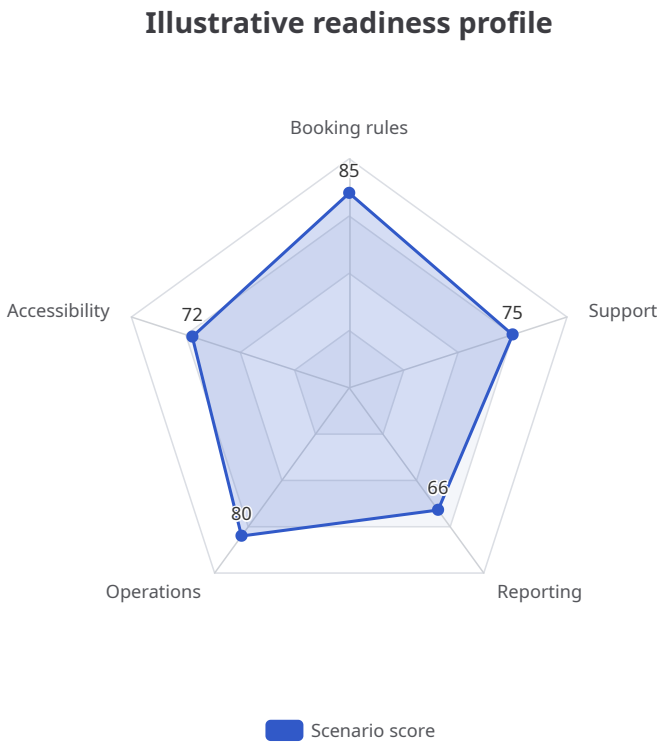
C10 — Donut chart: reservation channel

The ring view shows how members initiated bookings. Channel shares total 100%; they do not reuse the purpose breakdown.



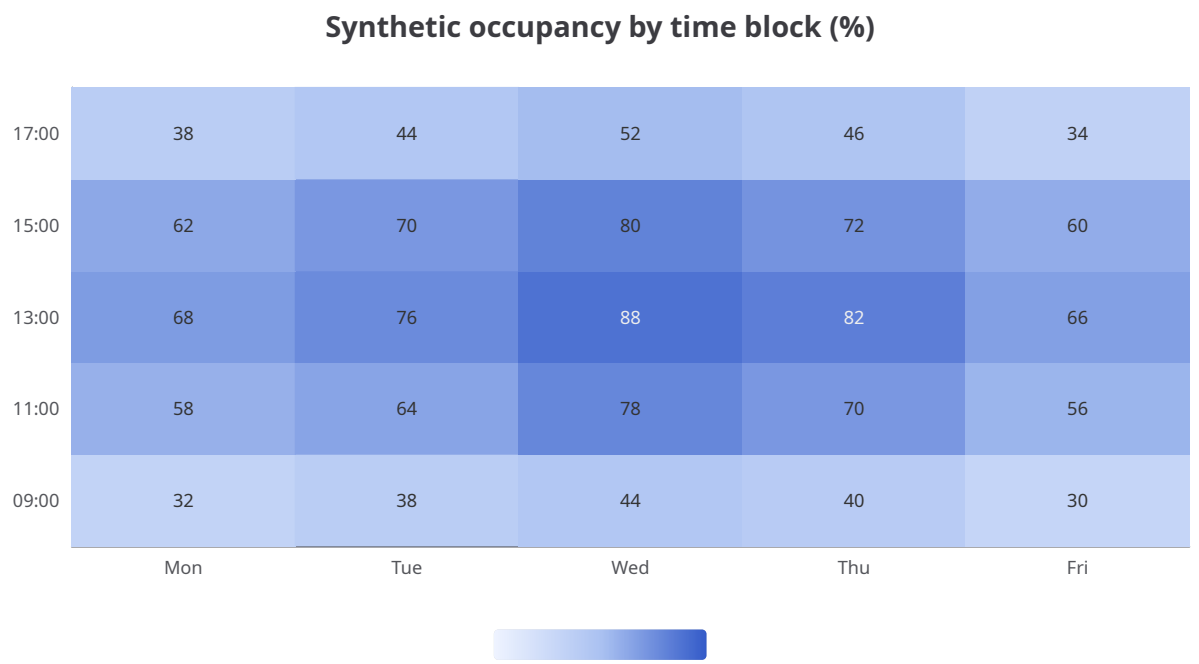
C11 — Radar chart: illustrative pilot readiness

Five sample checklist scores reveal uneven readiness. Scores are illustrative; they are not evidence that any feature exists.



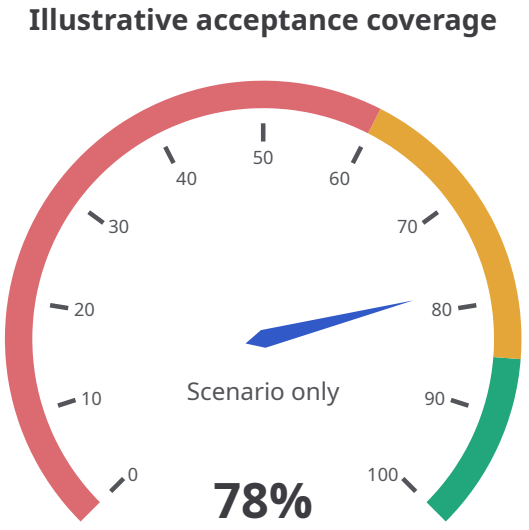
C12 — Heatmap: occupancy by weekday and time

Color shows the share of available capacity reserved within each time block. Cell labels repeat the values so color is not the only cue.



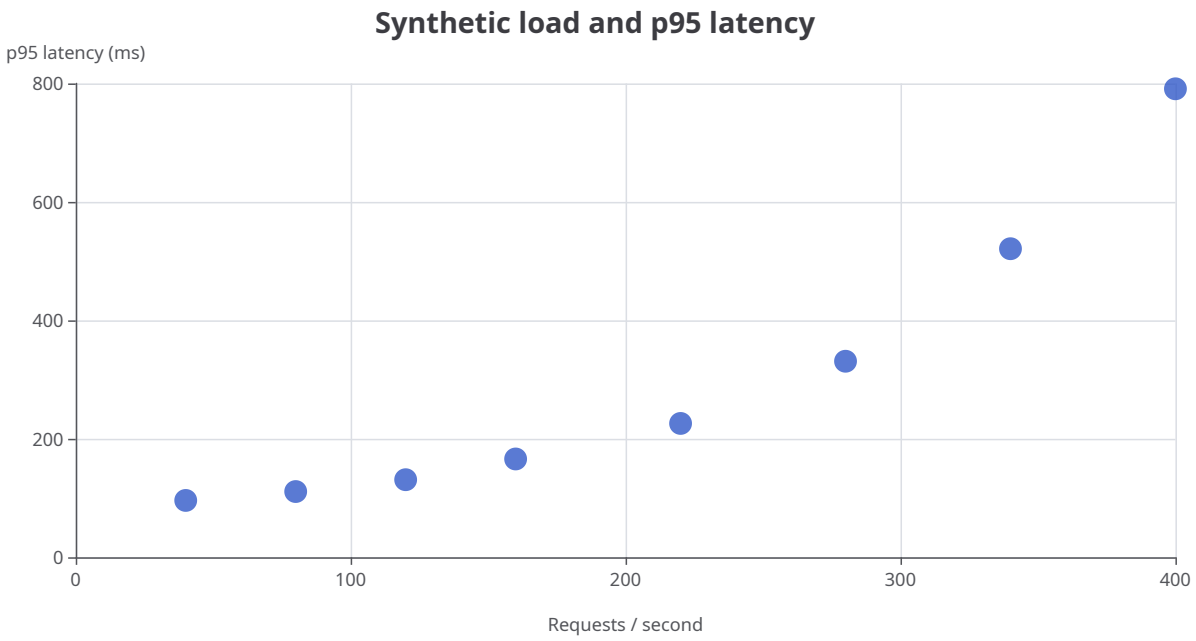
C13 — Gauge: example acceptance coverage

A single-value gauge demonstrates a hypothetical 78% result. A real release decision must use the acceptance evidence table, not this sample number.



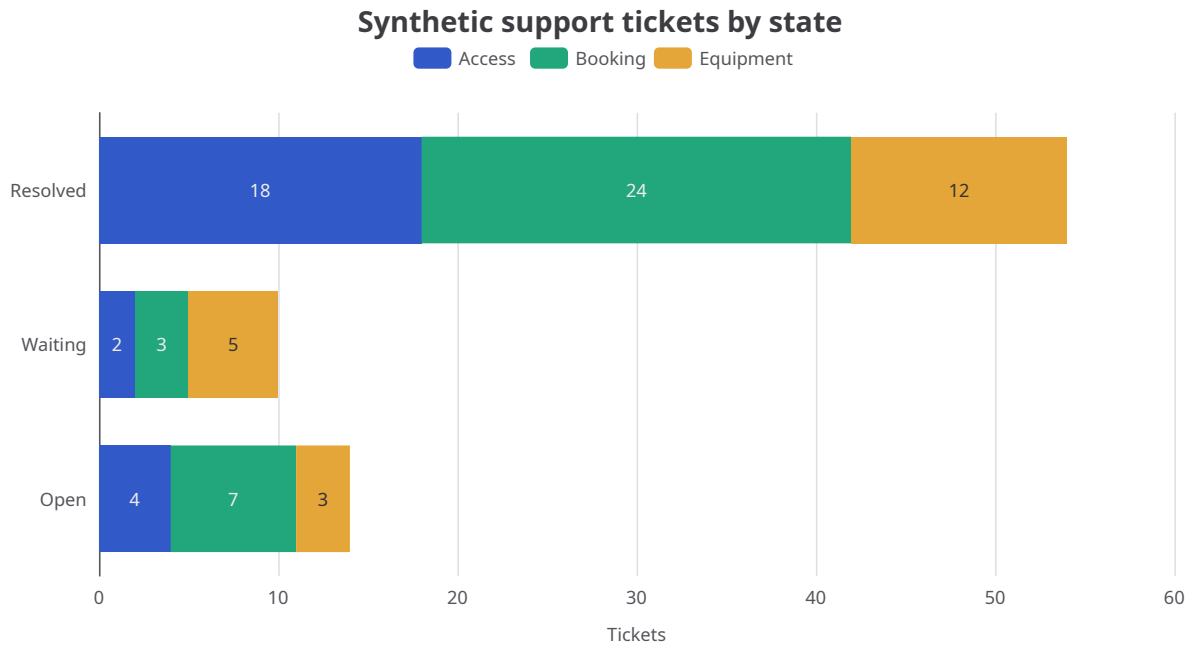
C14 — Scatter chart: load-test planning points

Each point is a fictional load-test scenario. This chart uses p95 per scenario; C06 uses individual request samples.



C15 — Stacked horizontal bars: support ticket mix

Horizontal stacks compare both total support volume and issue categories. These tickets are unrelated to booking lifecycle states.



5. Acceptance evidence to collect

These checks are planned work. No row claims a passing implementation result.

Area	Concrete check	Required evidence
Search	Change date or resource filters and read updated availability.	Browser session with keyboard-only navigation.
Concurrent booking	Submit the same interval from two sessions.	One committed booking; one 409; no duplicate event.
Idempotency	Repeat the same scoped key and payload.	The same booking ID; no extra reservation.
Approval expiry	Race a manager decision with the expiry worker.	Exactly one terminal decision; interval state remains consistent.
Workspace isolation	Use a valid member identity against another workspace.	No foreign booking, resource, or aggregate is exposed.
Time boundaries	Submit adjacent intervals and a daylight-saving transition.	Adjacent bookings succeed; ambiguous local input receives explicit feedback.
Check-in	Attempt check-in before, during, and after a confirmed interval.	Only the permitted transition is committed.

Area	Concrete check	Required evidence
Notifications	Retry a delivery after an acknowledgement is lost.	Stable event ID and recorded delivery outcome; no extra booking.
Accessibility	Complete search, conflict recovery, approval, and cancellation.	Visible focus, useful labels, status text, and readable error feedback.
Operations	Restore a backup and resume the queue in a rehearsal.	Verified records, replay results, and a recorded recovery duration.
Reporting	Recompute a small fixture by hand.	Chart totals match authorized source rows and stated denominators.

6. Open decisions

Question	Proposed starting point	Decision owner
Can managers approve their own requests?	Ask another manager to decide.	Product and workspace owner
What happens when no manager responds?	Expire after 30 minutes and release the interval.	Product and operations
Should a reminder use email, in-app, or both?	Start with email and an in-app status.	Product
How long should audit history remain available?	Define a retention period before the pilot.	Workspace owner
Should resource opening hours differ by location?	One schedule per resource in the first pilot.	Product
What is an acceptable latency and recovery target?	Agree thresholds after representative load and restore exercises.	Engineering and operations

7. Visual format coverage

The gallery uses 12 formats supported by the installed blueprint4llm profile. Aliases are not counted as additional formats.

Category	Format	Visual blocks	What it demonstrates
UI	Balsamiq Mockups 3	U01–U05	Search, empty, conflict, resource editor, and manager review states
UI	Pencil 2.21	U06	Positioned mobile details
UI	Figma JSON	U07–U08	Detailed confirmation and pending states
Diagrams	Mermaid	D01, D09–D12	Flowchart, state, sequence, Gantt, and ER diagrams
Diagrams	D2	D02–D04	Context, runtime containers, and SQL tables
Diagrams	Graphviz DOT	D05–D06	Dependencies and deployment clusters
Diagrams	draw.io JSON	D07	Grouped transaction and delivery graph
Diagrams	Excalidraw v2	D08	Positioned discovery map
Diagrams	PlantUML	D13–D15	Class, activity, and use-case diagrams
Diagrams	OpenAPI 3.1.1	D16	Operations, responses, and linked schemas
Charts	Vega-Lite 6	C01–C07	Bar, line, stacked area, scatter, histogram, box plot, and layered range
Charts	ECharts	C08–C15	Mixed, pie, donut, radar, heatmap, gauge, scatter, and stacked bars

Inventory: 39 visual blocks: 8 UI prototypes, 16 diagrams/API visuals, and 15 charts.

8. Review and source notes

Markdown is the source of truth. Keep one visual per fenced block so it can be inspected in isolation. Preserve the block language and metadata when rendering a review image.

Action	Command
Validate the plan	`blueprint4llm ./plan.md --check`
Start live preview	`blueprint4llm ./plan.md --serve --port=8811`

UI prototypes use Balsamiq Mockups 3, Pencil 2.21, and local Figma JSON. D2 label placement follows the D2 positioning guide. Renderer capabilities follow the installed blueprint4llm README. Figma blocks are local API-shaped JSON examples; they do not refer to an external design file. The API block is a proposed contract slice. No external assets or credentials are required.

HTML and PDF exports are optional review artifacts. Choose a format and output path before creating either export.